AS MATHEMATICS - PURE 1 - DAY 11

# Differentiation

Gradients, tangents and normals

YOUR AIM TODAY

## Differentiate powers and use the derivative as a gradient.

**YOUR RULE**

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

LEARN FROM ZERO

# The ideas you need

The derivative tells you how fast y changes as x changes.

- $d/dx(x^n)=n x^{(n-1)}$.
- Constants differentiate to 0.
- Differentiate every term separately.
- At x=a, substitute a into dy/dx for the tangent gradient.
- Normal gradient is the negative reciprocal of tangent gradient.

MEMORY RULE

Bring the power down, then reduce it by one.

## Derivative

WORKED STEPS

$y=3x^4-2x^2+7$ gives $dy/dx=12x^3-4x$.

## Tangent

WORKED STEPS

$y=x^2+1$ at $x=2$: point (2,5), gradient 4, so $y-5=4(x-2)$.

YOUR TURN

# Independent practice

Show every step. Include units in Mechanics.

| Question | Working and final answer |
|---|---|
| 1. Differentiate $x^5$. | |
| 2. Differentiate $4x^3-3x+8$. | |
| 3. Find gradient of $y=x^3-2x$ at $x=2$. | |
| 4. Find tangent to $y=x^2+3$ at $x=1$. | |
| 5. Find normal to $y=x^2$ at $x=2$. | |
| 6. If $y=2x^3+kx$ has gradient 13 at $x=1$, find k. | |

**STOP**

Do not continue until all six questions have an attempt.

ANSWER SECTION

# Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

| Q | Answer and essential working |
| --- | --- |
| 1 | 5x^4. |
| 2 | 12x^2-3. |
| 3 | dy/dx=3x^2-2; gradient 10. |
| 4 | Point (1,4), m=2: y=2x+2. |
| 5 | Tangent m=4, normal m=-1/4; point (2,4): y-4=-(1/4)(x-2). |
| 6 | dy/dx=6x^2+k; 6+k=13, so k=7. |

| Q | First wrong step | Correct rule |
| --- | --- | --- |
| | | |
| | | |
| | | |

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.